messaging Queue
Pub/sub
Queue
Kafka (mrp)
Rabbit MQ

Kafka is not just a MQ.

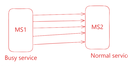
Messaging Queue

Benifits of Producer/Consumer

Producer and consumer can work on there
own pace.

Synchronous communication

Async Communication




HLD --> Throughput --> OPS (operations per second)

Throughput
Ola/uber/Rapido

freq = 1 signal/sec

Back of the envelope:

Delhi --> 1 Lakh driver in Ola
per sec --> 1 lakh req/sec

Add value to Db :
Fare calculation

SQL/ NoSQl (Casandra, Mongo)

1 Db Will have very high load.

Message Queues

1. Kafka

how kafka will solve above problem?

1 signal/second

Every DB that exist will not have such a high
throughput.

Kafka : Kafka is a MQ that have a very high
throughput.

Workers will do Bulk upload on DB.

JOb of a web server:
Take the request from client
Process the request
Put the data in DB.

Dbs are comparatively slower.
latency will be higher.

B+ trees/B trees

Kafka in depth

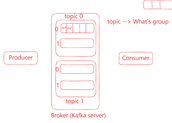
Why it is useful ?

1. Pace matching

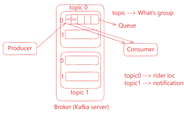
2. Async Communication (notification system)

In depth architecture of Kafka

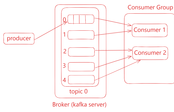
Cluster
broker
Topic
partition
Offset
Zookeper, kraft
Consumer
Consumer group
producer



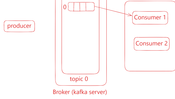

Introducing Producer-consumer Interaction




Consumer Groups &
How consumer actually consumes messages.

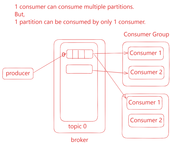



Auto-Balancing




If number of partitions are less then Number of
consumers, some consumers will remain idle.

1 consumer can consume multiple partitions.
But,
1 partition can be consumed by only 1 consumer.

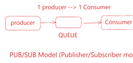



WHY kafka does all this ?

Consumer Group is fairly new in Kafka..

Messaging Queues serve 2 Purpose :
1. Queue (FIFO)
2. Pub/Sub

1 producer --> 1 Consumer

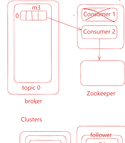



PUB/SUB Model (Publisher/Subscriber model)

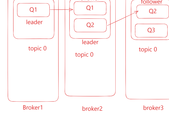
1 producer --> many consumer

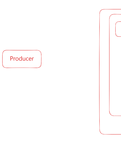
KAFKA --> LinkedIN

Zookeeper




Clusters




Leader --> Both Read & Write.
Follower --> Will just follow (listen) the leader